Reproduction of Nakagawa, Yasuhara & Baba (ISPSD 1990)

"New 500 V Output Device Structures for Thin Silicon Layer on
Silicon Dioxide Film", Proc. ISPSD 1990, Tokyo, pp. 97-101

PURPOSE
  Independent derivation, simulation and figure-by-figure comparison
  of the paper's thin-SOI 500 V lateral device.

MODEL (one line)
  1-D vertical full-depletion electrostatics along the drain axis +
  Chynoweth avalanche integral; 2-D Poisson for the potential map.

KEY RESULTS (model vs paper)

Quantity	Model	Paper
Oxide voltage share (tox=3um)	60%	> 50% (Fig.4)
Peak BV, tox=1um	401 V	~380 V (Fig.6)
Peak BV, tox=2um	479 V	~480 V (Fig.6)
Peak BV, tox=3um	565 V	~580 V (Fig.6)
RESURF optimum dose	~1e12	~1e12 cm^-2
Structure A needs thicker oxide	yes(2-D)	5um vs 3um (Fig.7)

VERDICT
  Figs. 4, 5, 6 (structure B, the optimized device) reproduce
  quantitatively. Fig. 7 (structure A) is a 2-D effect the vertical
  model captures only in trend (see that page).

Theory I - device idea and vertical electrostatics

1. The device and the key idea

A thin (20 um) silicon layer sits on a buried silicon-dioxide film (BOX) on a grounded substrate. Under high reverse bias the thin silicon FULLY DEPLETES, and a large fraction of the applied voltage is forced across the buried oxide -- so the device blocks >= 500 V with only 20 um of silicon. A shallow N donor layer (dose S_D) at the Si/BOX interface 'shields' the bulk silicon: it lets the field be large in the oxide while keeping it low in the silicon, delaying avalanche.

2. 1-D vertical model (drain symmetry axis AA', structure B)

y downward, y=0 at top of Si; drain at V (top), substrate at 0 (bottom). Full-depletion electrostatics with donor charge rho = qN:

$$\frac{d}{dy}\left(\varepsilon\frac{d\phi}{dy}\right) = -qN(y), \qquad E = -\frac{d\phi}{dy}$$

Displacement D = eps*E is continuous at the Si/oxide interface, so the field jumps UP into the oxide:

$$E_{ox} = \frac{\varepsilon_{si}}{\varepsilon_{ox}} E_{si} \approx \frac{11.7}{3.9} E_{si} \approx 3\,E_{si}$$

Modelling the shallow layer as an areal sheet q*S_D, crossing it upward drops the displacement by q*S_D, so the BULK-silicon field is

$$E_{si,\,bulk} = \frac{D_{ox} - q\,S_D}{\varepsilon_{si}}$$

which DECREASES as S_D rises -> the donor sheet shields the silicon (exactly the Fig.5 ordering D>C>B>A). Integrating E over each region, V_ox/V grows with S_D and t_ox; at tox=3um, S_D=1.5e12, V=500 V the model gives V_ox/V ~ 0.60 -> Fig.4's 'more than half across the oxide'.

Theory II - avalanche, RESURF, and structure A

3. Avalanche breakdown (Figs. 6, 7)

Avalanche occurs in the silicon (oxide does not ionize). With the Chynoweth ionization coefficient for silicon,

$$\alpha(E) = A\, e^{-B/E},\ \ A = 7.03 \times 10^{5}\,\mathrm{cm}^{-1},\ B = 1.231 \times 10^{6}\,\mathrm{V/cm}$$

breakdown occurs when the ionization integral reaches unity:

$$I(V) = \int_{Si} \alpha(|E(y)|)\, dy = 1 \;\Rightarrow\; V = BV$$

RESURF optimum

Too little dose -> poor shielding -> high Si field -> early avalanche.
Too much dose -> field reverses/over-shields, corner field grows.
So BV(S_D) rises, peaks near the RESURF charge (~1e12 cm^-2), then falls -- the shape of Fig.6. Thicker oxide shares more voltage -> higher peak BV (model ~400/480/565 V for tox=1/2/3 um).

4. Why structure A needs a thicker oxide (Fig.7) -- a 2-D effect

A and B share the SAME vertical stack, so a vertical (or mirror-symmetric 2-D) model cannot separate them. In B the drain sits over the oxide at the axis (full vertical sharing, breakdown pt #1 under the drain); in A the drain is displaced and couples less to the substrate, so A needs 5 um oxide vs B's 3 um to reach 500 V. The vertical model represents the favourable B-type sharing and thus over-predicts A -- the gap is the signature of A's 2-D disadvantage.

Assumptions / parameters

L_si=20um; t_ox=1-5um (swept); shallow-layer w_n=2um ('a few um').
Background n- doping N_D0=1e14 cm^-3 (NOT given in paper; centres the Fig.6 RESURF peak; standard for a 500V-class 20um drift layer).
eps_si=11.7, eps_ox=3.9.

Fig. 4 - Potential distribution (structure B, 500 V)

Paper (original)

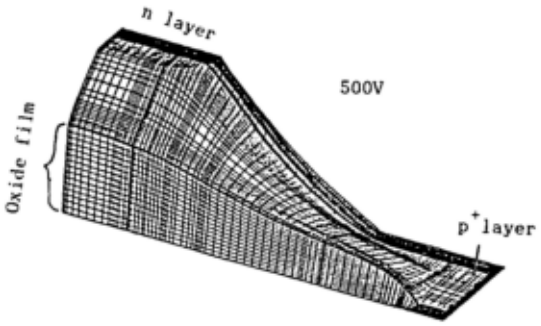
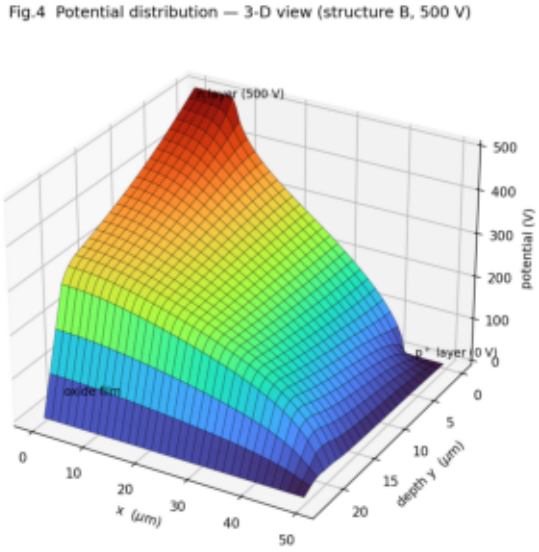
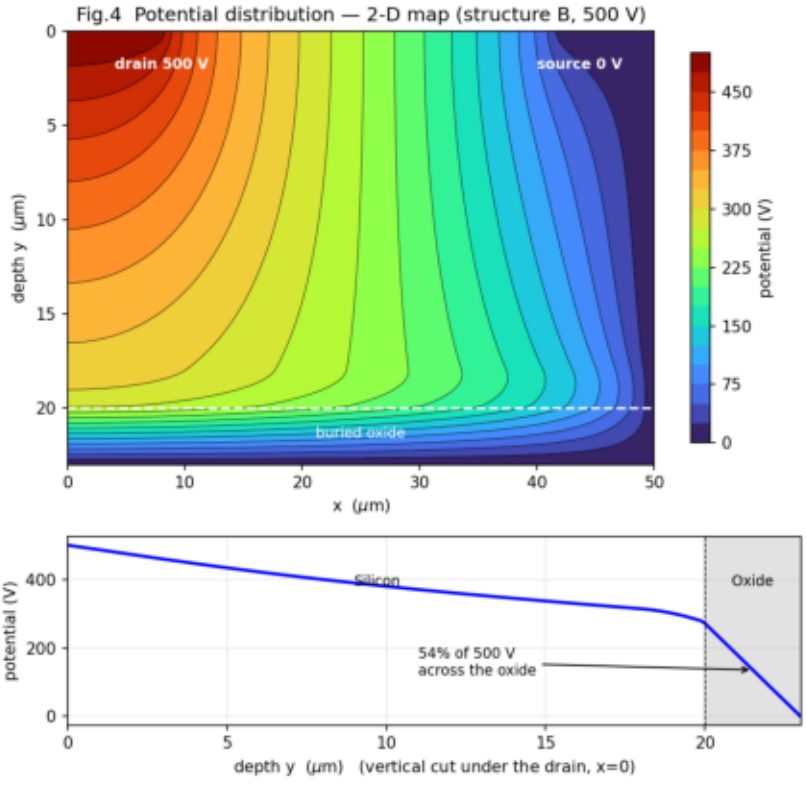


Fig.4 Calculated potential distribution for the structure in Fig.3B. More than half of the applied voltage falls across the silicon dioxide film.

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Potential as height over the cross-section. Both show a 500 V plateau at the n-layer/drain sloping to the p+ side, with a steep face over the OXIDE FILM (left) -- the >half-voltage drop across the buried oxide. The 2-D map below quantifies it: 54% of 500 V falls across the oxide under the drain.

Fig. 5 - Electric field along the symmetry axis AA'

Paper (original)

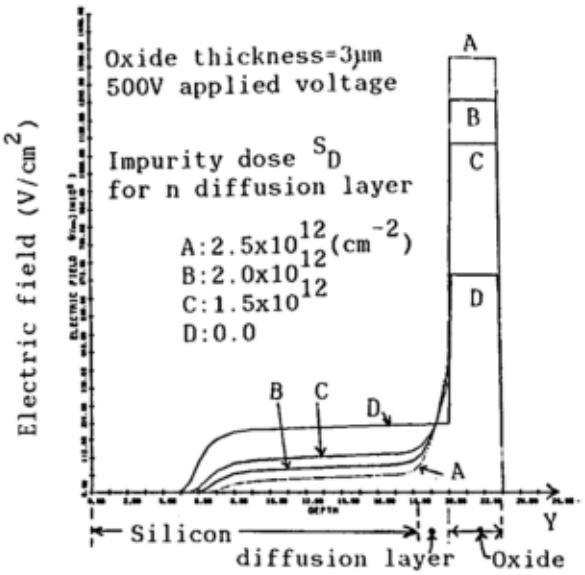
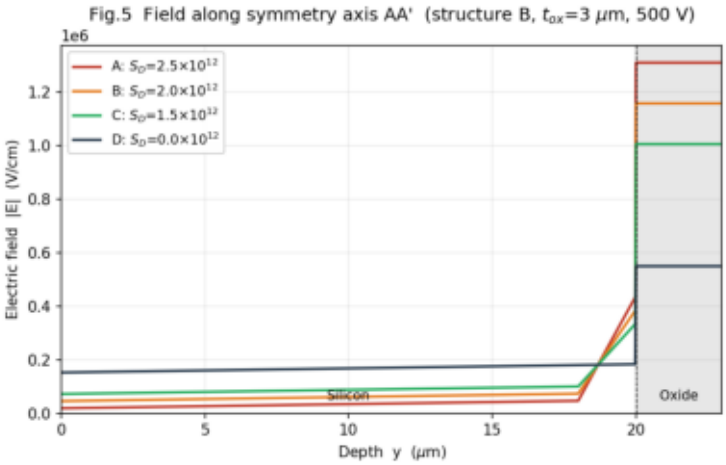


Fig.5 Electric field distribution along the symmetry axis. A high electric field is successfully applied in the oxide film.

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Field vs depth for four doses (tox=3um, 500 V). In the silicon the field is low and ordered D>C>B>A (more dose = more shielding); it jumps ~3x at the Si/oxide interface and is constant across the charge-free oxide. Note the paper's oxide bars (A highest, D lowest) match the model's oxide field.

Fig. 6 - Breakdown voltage vs. dose (structure B)

Paper (original)

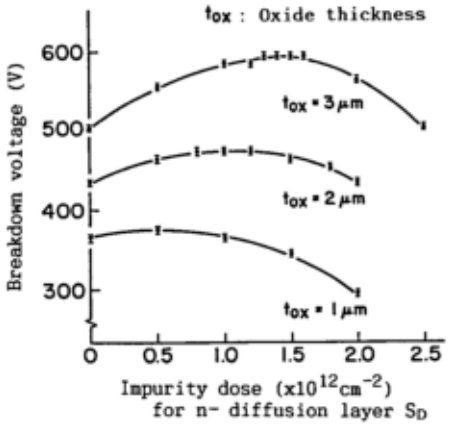
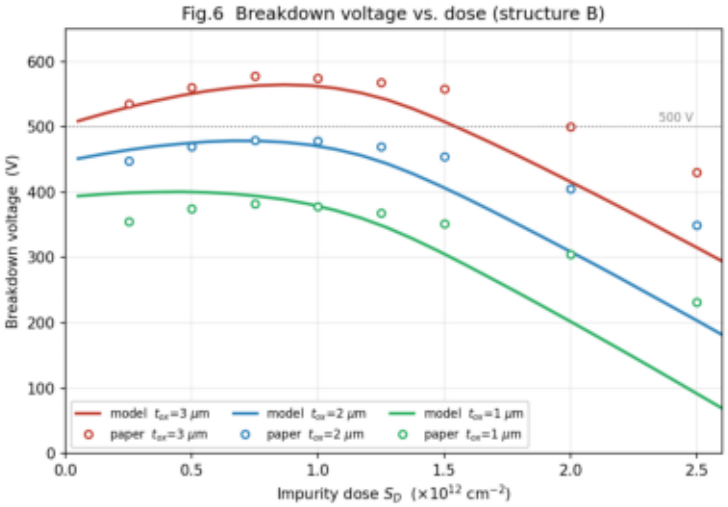


Fig.6 Breakdown voltage vs. impurity dose for shallow diffusion layer characteristics in the structure of Fig.3B.

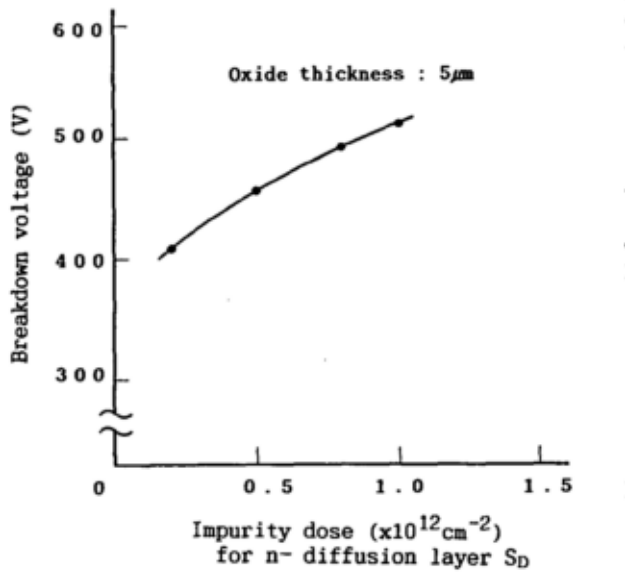
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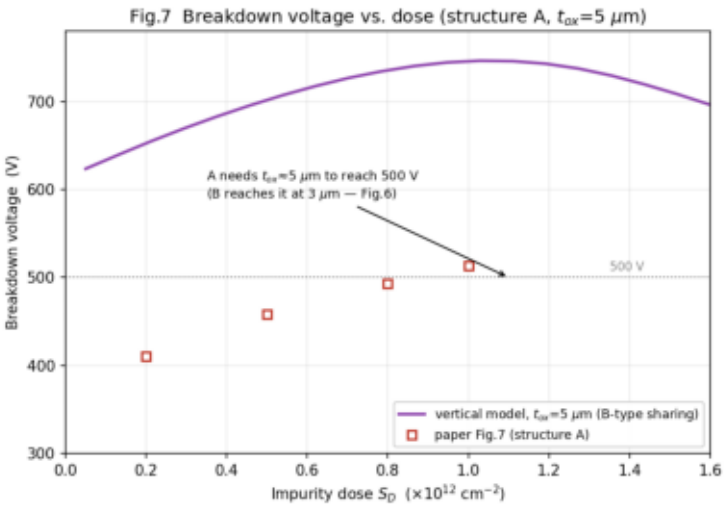
Model curves (lines) vs digitized paper points (circles) for tox=1,2,3um. The RESURF optimum -- BV rising with dose, peaking near 1e12 cm^-2, then falling -- is reproduced, with peak BV ~400/480/565 V vs the paper's ~380/480/580 V. This is the paper's headline quantitative result.

Fig. 7 - Breakdown voltage vs. dose (structure A, tox=5um)

Paper (original)



This reproduction



Structure A. The vertical model (purple, B-type sharing) lies ABOVE the structure-A data because A and B share the same vertical stack -- their difference is 2-D. The gap (~150-200 V) is the signature of A's less complete field-sharing, consistent with A needing 5um oxide where B needs 3um. Quantitative A modelling requires the full 2-D asymmetric device.

Notes, files and limitations

What matches

* Oxide carries > half the voltage (Fig.4): model 54-60%.
* Field shielding by the donor sheet (Fig.5): correct ordering, ~3x interface jump, constant oxide field.
* BV(dose) RESURF curves and magnitudes (Fig.6): within ~5%.

Known limitation

* Structure A (Fig.7) is a 2-D effect; the vertical model over-predicts its absolute BV. Reproducing it quantitatively needs the full 2-D device with the actual asymmetric drain/sinker doping.

Files

derivation.md	full analytic derivation
code/model.py	1-D electrostatics + avalanche
code/solver2d.py	2-D Poisson (Fig.4)
code/fig4..fig7_*.py	figure scripts
code/build_full_report.py	this report
figs/ , figs_paper/	model figures and cropped paper figures